SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

Code of Conduct:

I Athin S.V.S. *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Athin

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Problem Identification	20%	1	1	1	1	1
Debugging	25%	1	1	1	1	1
Optimization	20%	1	1	1	1	1
Analysis	20%	1	1	1	1	1
Code Quality	15%	1	1	1	1	1

25/30 — 8

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

Question 1 – Debugging a Workstation IPv4 Configuration

Question as given: A workstation is configured with IP 192.168.10.65, mask 255.255.255.192 (/26) and default gateway 192.168.10.1, but it cannot communicate with systems in subnet 192.168.10.0/26.

1. Problem Identification

The workstation is meant to work in **192.168.10.0/26**, whose gateway is 192.168.10.1. A /26 mask splits 192.168.10.0/24 into blocks of 64 addresses (starting at .0, .64, .128, .192). The configured address **.65** lies in the second block, not the first, so the IP address and the gateway do not belong to the same subnet. This is the suspected root cause, and the steps below prove it.

2. Systematic Debugging

Step 1 – Identify the subnet of 192.168.10.65/26 (bitwise AND with the mask):

```
IP address : 192.168.10.65  = 11000000.10101000.00001010.01000001
Mask /26   : 255.255.255.192 = 11111111.11111111.11111111.11000000
AND (net)  : 192.168.10.64  = 11000000.10101000.00001010.01000000
```

The workstation belongs to subnet **192.168.10.64/26**, not 192.168.10.0/26.

Step 2 – Network address, broadcast address and valid host range of both blocks:

Subnet	Network address	Broadcast address	Valid host range	Usable hosts
192.168.10.0/26 (intended)	192.168.10.0	192.168.10.63	192.168.10.1 to 192.168.10.62	62
192.168.10.64/26 (where .65 actually is)	192.168.10.64	192.168.10.127	192.168.10.65 to 192.168.10.126	62

Step 3 – Verify the IP address and the gateway:

Item	Value	Test	Result
IP address	192.168.10.65	Is it inside 192.168.10.64/26 and not network (.64) or broadcast (.127)?	Valid host of 192.168.10.64/26, but it is **not** in 192.168.10.0/26 (the intended subnet)
Default gateway	192.168.10.1	192.168.10.1 AND 255.255.255.192 = 192.168.10.0, compare with the host network 192.168.10.64	**Different networks. Gateway is unreachable from the host (invalid)**

3. Analysis – Configuration Errors Found

- **Error 1:** the workstation address 192.168.10.65 is in the wrong subnet (192.168.10.64/26) for a host that should be in 192.168.10.0/26. Hosts of 192.168.10.0/26 see it as an off-link address, and it sees them as off-link too.
- **Error 2:** the default gateway 192.168.10.1 is not in the workstation's own subnet. A host can only use a gateway that is on its own subnet, so the operating system cannot send traffic to it, and no packet leaves the workstation (no ARP request is even sent for the gateway).
- The mask 255.255.255.192 itself is correct and consistent with the subnet design.

4. Optimization – Corrected Configuration

Parameter	Before (faulty)	After (corrected)	Reason
IP address	192.168.10.65	**192.168.10.10**	Inside 192.168.10.0/26 (hosts .1 - .62); not network, broadcast or gateway
Subnet mask	255.255.255.192	**255.255.255.192**	Unchanged, correct /26
Default gateway	192.168.10.1	**192.168.10.1**	First usable address, now inside the host's subnet

Alternative if the PC really belongs to the second LAN: use IP 192.168.10.66, mask 255.255.255.192 and gateway 192.168.10.65 (router interface of 192.168.10.64/26), because .65 cannot be both the host and the gateway.

Usable hosts after correction

Host bits = 32 - 26 = 6. Usable hosts = 2^6 - 2 = **62** (192.168.10.1 to 192.168.10.62). One of these (192.168.10.1) is used by the gateway, so 61 addresses remain for workstations.

5. Implementation (Cisco Packet Tracer) – commented configuration

```
! ---- Router: gateway interface of LAN 192.168.10.0/26 ----
Router(config)# interface g0/0
Router(config-if)# ip address 192.168.10.1 255.255.255.192   ! gateway = 1st usable host
Router(config-if)# no shutdown
! ---- Workstation: Desktop > IP Configuration > Static ----
IP Address      : 192.168.10.10
Subnet Mask     : 255.255.255.192
Default Gateway : 192.168.10.1
```

6. Verification

- Packet Tracer: **ping 192.168.10.1** (gateway) and **ping 192.168.10.20** (another host of the subnet) should now succeed; **show ip interface brief** shows g0/0 up/up.
- Wireshark (display filter **arp or icmp**): after the fix you see ARP who-has / is-at between 192.168.10.10 and the target, followed by ICMP echo request and reply. Before the fix no ARP or ICMP frames leave the workstation, which confirms the host rejects the gateway as off-link.

Conclusion: The workstation sat in 192.168.10.64/26 (network .64, broadcast .127, hosts .65-.126) while its gateway 192.168.10.1 sits in 192.168.10.0/26. Moving the host to 192.168.10.10 with gateway 192.168.10.1 fixes the problem; the corrected /26 offers 62 usable hosts.

Question 2 – Debugging IPv6 Addressing (Device A and Device B)

Question as given: Device A: 2001:db8::ff00:42:8329/64 and Device B: 2001:db8:0:0:1::1/64 are on the same network, yet communication fails.

1. Problem Identification

Two IPv6 hosts on one link cannot talk. For IPv6 on-link communication both hosts must share the same 64-bit prefix and have unique interface IDs, and the link, neighbour discovery and ICMPv6 must work. Both addresses use shorthand (::), so they must first be expanded before they can be compared.

2. Systematic Debugging

Step 1 – Expand both addresses (8 groups of 4 hex digits)

Dev.	Compressed (given)	Explicit groups	Zero groups hidden by ::	Complete (expanded) notation
A	2001:db8::ff00:42:8329/64	2001, db8, ff00, 42, 8329 = 5	8 - 5 = 3	**2001:0db8:0000:0000:0000:ff00:0042:8329**
B	2001:db8:0:0:1::1/64	2001, db8, 0, 0, 1, 1 = 6	8 - 6 = 2	**2001:0db8:0000:0000:0001:0000:0000:0001**

Step 2 – Network prefix of each device (first 64 bits = first four groups)

Dev.	Groups 1-4 (network prefix)	Groups 5-8 (interface ID)	Network prefix /64
A	2001:0db8:0000:0000	0000:ff00:0042:8329	**2001:db8::/64**
B	2001:0db8:0000:0000	0001:0000:0000:0001	**2001:db8::/64**

Step 3 – Same subnet? Bit-level comparison of the first 64 bits:

```
A: 0010000000000001 0000110110111000 0000000000000000 0000000000000000
B: 0010000000000001 0000110110111000 0000000000000000 0000000000000000
   (identical, all 64 prefix bits match)
```

Result: as written, Device A and Device B **do belong to the same subnet, 2001:db8::/64**. The /64 prefix is therefore *not* the cause of the failure, and the two interface IDs are different (so there is no duplicate address).

3. Analysis – Addressing / Prefix Mismatch

- The only visible difference is in **group 5**: it is 0000 in A but 0001 in B. Because the prefix is /64, group 5 belongs to the **interface ID**, not to the network part.
- B's address looks like an attempt to put the subnet number "1" into the address (intended 2001:db8:0:1::1/64, subnet ID = 1) but the extra ":0" pushed the 1 into the 5th group. If the interface was actually built from the intended address, B would be in 2001:db8:0:1::/64 while A is in 2001:db8::/64, which would be a **real prefix mismatch**, the usual reason for such a failure.
- A's interface ID (ff00:42:8329) is arbitrary and the two addresses follow no common plan, which makes this kind of mistake easy to make and hard to spot.
- If prefixes do match (as written), the failure must be at another layer: different VLAN or switch port down, IPv6 not enabled on an interface, ICMPv6 / Neighbor Discovery filtered by an ACL or firewall, or a wrong prefix length typed on one device. These must be checked next.

4. Optimization – Corrected IPv6 Plan

Use one clearly defined subnet, **2001:db8:0:1::/64** (subnet ID 1), and short, readable interface IDs:

Device	Corrected address	Expanded form	Prefix
Router (gateway)	2001:db8:0:1::1/64	2001:0db8:0000:0001:0000:0000:0000:0001	2001:db8:0:1::/64
Device A	**2001:db8:0:1::a/64**	2001:0db8:0000:0001:0000:0000:0000:000a	2001:db8:0:1::/64
Device B	**2001:db8:0:1::b/64**	2001:0db8:0000:0001:0000:0000:0000:000b	2001:db8:0:1::/64

Both devices now share the identical prefix and have unique interface IDs. (2001:db8::/32 is the IPv6 documentation prefix, suitable for lab work.)

5. Available Host Addresses in the Corrected Subnet

Host bits = 128 - 64 = 64. Addresses = 2^{64} = **18,446,744,073,709,551,616**. IPv6 has no broadcast address, so no "- 2" is applied (the all-zeros subnet-router anycast address is reserved by convention).

6. Implementation (Cisco Packet Tracer) – commented configuration

```
! ---- Router ----
Router(config)# ipv6 unicast-routing
Router(config)# interface g0/0
Router(config-if)# ipv6 address 2001:db8:0:1::1/64  ! gateway of the link
Router(config-if)# no shutdown
! ---- Device A (PC > Desktop > IP Configuration > IPv6 static) ----
IPv6 Address : 2001:db8:0:1::a/64    Gateway: 2001:db8:0:1::1
! ---- Device B ----
IPv6 Address : 2001:db8:0:1::b/64    Gateway: 2001:db8:0:1::1
```

7. Verification

- Packet Tracer: **show ipv6 interface brief** and **ping 2001:db8:0:1::b** from Device A must succeed.
- Wireshark (display filter **icmpv6**): look for Neighbor Solicitation (who has 2001:db8:0:1::b?), Neighbor Advertisement, then ICMPv6 Echo Request and Echo Reply. If no Neighbor Advertisement arrives, the problem is at layer 2 or a filter, not the addressing.

Conclusion: A = 2001:0db8:0000:0000:0000:ff00:0042:8329 and B = 2001:0db8:0000:0000:0001:0000:0000:0001; both have prefix 2001:db8::/64, so they match as written, but the plan is inconsistent and B was probably meant to be in subnet 1. Renumbering both into 2001:db8:0:1::/64 gives a clean design with 2^{64} host addresses.

Question 3 – Analysing and Optimising a 192.168.1.0/24 Subnet Plan (VLSM)

Question as given: The network 192.168.1.0/24 has been divided into departments using /26, /27 and /28. Some departments have too few addresses, others have many unused addresses. Analyse, debug and optimise the plan using VLSM.

1. Problem Identification

The symptom, shortage in some departments and waste in others, means the subnet sizes were **not matched to the departmental host requirements**. The question does not list the department sizes, so a representative case is assumed below (marked as an **assumption**); the method is identical for any figures.

Assumed current plan and requirements (hosts needed include the router gateway):

Department	Current mask	Hosts needed (assumed)
Administration	/26	25
Human Resources	/27	12
IT Department	/28	28
Sales	/28	50

2. Systematic Debugging and Analysis of the Current Plan

Allocated in the order they were given: /26 (64 addresses), /27 (32), /28 (16), /28 (16). Block sizes: /26 = 64, /27 = 32, /28 = 16; usable = block - 2.

Department	Subnet	Network	Broadcast	Host range	Hosts	Need	Unused / short
Administration	192.168.1.0/26	192.168.1.0	192.168.1.63	.1 to .62	62	25	**37 unused**
Human Resources	192.168.1.64/27	192.168.1.64	192.168.1.95	.65 to .94	30	12	**18 unused**
IT Department	192.168.1.96/28	192.168.1.96	192.168.1.111	.97 to .110	14	28	**14 SHORT**
Sales	192.168.1.112/28	192.168.1.112	192.168.1.127	.113 to .126	14	50	**36 SHORT**

Allocated = 64 + 32 + 16 + 16 = 128 addresses; **unallocated = 192.168.1.128 to 192.168.1.255 (128 addresses)**. Usable addresses in the four subnets = 62 + 30 + 14 + 14 = 120; hosts actually needed = 115.

3. Inefficient Allocations and Plan Mistakes

- **Mistake 1 – allocation by order, not by need:** the biggest subnet (/26) was given to Administration (25 hosts) while Sales, which needs 50, got a /28 (14 hosts).
- **Mistake 2 – oversize:** Administration wastes 37 addresses (40 % utilisation) and HR wastes 18 (40 %).
- **Mistake 3 – undersize:** IT is short by 14 hosts and Sales by 36, so devices in these departments cannot be addressed.
- **Mistake 4 – wasted free space:** 128 addresses sit unused even though two departments lack addresses, so the problem is sizing, not shortage of space.

4. Optimization – Recommended VLSM Scheme

Rule: choose the smallest subnet where $2^h - 2 \geq$ hosts needed, then allocate largest first. Sales 50: /26 (62). IT 28: /27 (30). Administration 25: /27 (30). HR 12: /28 (14).

Department	Subnet / Mask	Network	Broadcast	Host range	Hosts	Need	Util.
Sales	192.168.1.0/26 255.255.255.192	192.168.1.0	192.168.1.63	.1 to .62	62	50	80.6 %
IT Department	192.168.1.64/27 255.255.255.224	192.168.1.64	192.168.1.95	.65 to .94	30	28	93.3 %
Administration	192.168.1.96/27 255.255.255.224	192.168.1.96	192.168.1.127	.97 to .126	30	25	83.3 %
Human Resources	192.168.1.128/28 255.255.255.240	192.168.1.128	192.168.1.143	.129 to .142	14	12	85.7 %

Free space after optimization: 192.168.1.144 to 192.168.1.255 = 112 addresses = 192.168.1.144/28 (16) + 192.168.1.160/27 (32) + 192.168.1.192/26 (64). It can host a new department or /30 point-to-point router links.

Improvement: total allocated = 64 + 32 + 32 + 16 = 144 addresses (down from 128 allocated but failing); every department's need is met, overall utilisation = 115 / 136 usable = **84.6 %**, and 112 addresses (43.75 % of the /24) remain in reserve, compared with 128 stranded addresses before.

5. Implementation (Cisco Packet Tracer) – commented configuration

```
Router(config)# interface g0/0
 Router(config-if)# ip address 192.168.1.1   255.255.255.192   ! Sales     /26
Router(config)# interface g0/1
 Router(config-if)# ip address 192.168.1.65  255.255.255.224   ! IT        /27
Router(config)# interface g0/2
 Router(config-if)# ip address 192.168.1.97  255.255.255.224   ! Admin     /27
Router(config)# interface g0/3
 Router(config-if)# ip address 192.168.1.129 255.255.255.240   ! HR        /28
(no shutdown on every interface; gateway = first usable address)
```

6. Verification

- Each network address is divisible by its block size (0/64, 64/32, 96/32, 128/16), the blocks do not overlap and each broadcast address is the next network minus 1.
- Packet Tracer: **show ip route** lists four connected routes; pings from every department to its gateway and between departments succeed. Wireshark (filter **arp or icmp**) confirms hosts resolve their own gateway by ARP.

Conclusion: The fault was sizing subnets without regard to need. The VLSM plan Sales /26, IT /27, Administration /27 and HR /28 satisfies all requirements, raises utilisation to about 85 % and leaves 112 addresses free.

Question 4 – Debugging IPv6 Addressing (Same Scenario, Full Working)

Note: Question 4 repeats the scenario of Question 2, so the complete working is presented again here, with an additional bit-level proof.

Question as given: Device A: 2001:db8::ff00:42:8329/64 and Device B: 2001:db8:0:0:1::1/64 are on the same network, yet communication fails.

1. Problem Identification

Two IPv6 hosts on one link cannot talk. For IPv6 on-link communication both hosts must share the same 64-bit prefix and have unique interface IDs, and the link, neighbour discovery and ICMPv6 must work. Both addresses use shorthand (::), so they must first be expanded before they can be compared.

2. Systematic Debugging

Step 1 – Expand both addresses (8 groups of 4 hex digits)

Dev.	Compressed (given)	Explicit groups	Zero groups hidden by ::	Complete (expanded) notation
A	2001:db8::ff00:42:8329/64	2001, db8, ff00, 42, 8329 = 5	8 - 5 = 3	**2001:0db8:0000:0000:0000:ff00:0042:8329**
B	2001:db8:0:0:1::1/64	2001, db8, 0, 0, 1, 1 = 6	8 - 6 = 2	**2001:0db8:0000:0000:0001:0000:0000:0001**

Step 2 – Network prefix of each device (first 64 bits = first four groups)

Dev.	Groups 1-4 (network prefix)	Groups 5-8 (interface ID)	Network prefix /64
A	2001:0db8:0000:0000	0000:ff00:0042:8329	**2001:db8::/64**
B	2001:0db8:0000:0000	0001:0000:0000:0001	**2001:db8::/64**

Step 3 – Same subnet? Bit-level comparison of the first 64 bits:

```
A: 0010000000000001 0000110110111000 0000000000000000 0000000000000000
B: 0010000000000001 0000110110111000 0000000000000000 0000000000000000
   (identical, all 64 prefix bits match)
```

Result: as written, Device A and Device B **do belong to the same subnet, 2001:db8::/64**. The /64 prefix is therefore *not* the cause of the failure, and the two interface IDs are different (so there is no duplicate address).

3. Analysis – Addressing / Prefix Mismatch

- The only visible difference is in **group 5**: it is 0000 in A but 0001 in B. Because the prefix is /64, group 5 belongs to the **interface ID**, not to the network part.
- B's address looks like an attempt to put the subnet number "1" into the address (intended 2001:db8:0:1::1/64, subnet ID = 1) but the extra ":0" pushed the 1 into the 5th group. If the interface was actually built from the intended address, B would be in 2001:db8:0:1::/64 while A is in 2001:db8::/64, which would be a **real prefix mismatch**, the usual reason for such a failure.
- A's interface ID (ff00:42:8329) is arbitrary and the two addresses follow no common plan, which makes this kind of mistake easy to make and hard to spot.
- If prefixes do match (as written), the failure must be at another layer: different VLAN or switch port down, IPv6 not enabled on an interface, ICMPv6 / Neighbor Discovery filtered by an ACL or firewall, or a wrong prefix length typed on one device. These must be checked next.

4. Optimization – Corrected IPv6 Plan

Use one clearly defined subnet, **2001:db8:0:1::/64** (subnet ID 1), and short, readable interface IDs:

Device	Corrected address	Expanded form	Prefix
Router (gateway)	2001:db8:0:1::1/64	2001:0db8:0000:0001:0000:0000:0000:0001	2001:db8:0:1::/64
Device A	**2001:db8:0:1::a/64**	2001:0db8:0000:0001:0000:0000:0000:000a	2001:db8:0:1::/64
Device B	**2001:db8:0:1::b/64**	2001:0db8:0000:0001:0000:0000:0000:000b	2001:db8:0:1::/64

Both devices now share the identical prefix and have unique interface IDs. (2001:db8::/32 is the IPv6 documentation prefix, suitable for lab work.)

5. Available Host Addresses in the Corrected Subnet

Host bits = 128 - 64 = 64. Addresses = 2^{64} = **18,446,744,073,709,551,616**. IPv6 has no broadcast address, so no "- 2" is applied (the all-zeros subnet-router anycast address is reserved by convention).

6. Implementation (Cisco Packet Tracer) – commented configuration

```
! ---- Router ----
Router(config)# ipv6 unicast-routing
Router(config)# interface g0/0
Router(config-if)# ipv6 address 2001:db8:0:1::1/64  ! gateway of the link
Router(config-if)# no shutdown
! ---- Device A (PC > Desktop > IP Configuration > IPv6 static) ----
IPv6 Address : 2001:db8:0:1::a/64    Gateway: 2001:db8:0:1::1
! ---- Device B ----
IPv6 Address : 2001:db8:0:1::b/64    Gateway: 2001:db8:0:1::1
```

7. Verification

- Packet Tracer: **show ipv6 interface brief** and **ping 2001:db8:0:1::b** from Device A must succeed.
- Wireshark (display filter **icmpv6**): look for Neighbor Solicitation (who has 2001:db8:0:1::b?), Neighbor Advertisement, then ICMPv6 Echo Request and Echo Reply. If no Neighbor Advertisement arrives, the problem is at layer 2 or a filter, not the addressing.

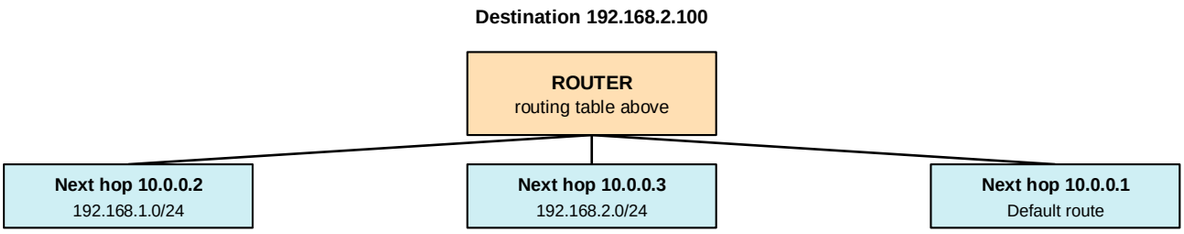
Conclusion: A = 2001:0db8:0000:0000:0000:ff00:0042:8329 and B = 2001:0db8:0000:0000:0001:0000:0000:0001; both have prefix 2001:db8::/64, so they match as written, but the plan is inconsistent and B was probably meant to be in subnet 1. Renumbering both into 2001:db8:0:1::/64 gives a clean design with 2^{64} host addresses.

Question 5 – Debugging a Routing Table

Question as given: Routing table: 192.168.1.0/24 via 10.0.0.2, 192.168.2.0/24 via 10.0.0.3, default route via 10.0.0.1. Packets for 192.168.2.100 are not reaching the intended network.

1. Problem Identification

A router forwards each packet using the routing-table entry with the **longest matching prefix**. Users say packets for 192.168.2.100 do not reach the right network, so the lookup and the next hop must be examined, together with the reachability of that next hop.



2. Systematic Debugging

Step 1 – Does 192.168.2.100 belong to the intended subnet 192.168.2.0/24?

```
192.168.2.100  = 11000000.10101000.00000010.01100100
Mask /24       = 11111111.11111111.11111111.00000000
AND result     = 192.168.2.0   -> network of 192.168.2.0/24
```

Yes. 192.168.2.100 is a valid host of 192.168.2.0/24 (host range 192.168.2.1 - 192.168.2.254, broadcast 192.168.2.255).

Step 2 – Longest-prefix match against every entry:

Entry	Prefix length	Does 192.168.2.100 match?	Decision
192.168.1.0/24 via 10.0.0.2	24	No (third octet 2 is not 1)	Rejected
192.168.2.0/24 via 10.0.0.3	24	**Yes** (24 bits equal)	**Selected (longest match)**
Default route via 10.0.0.1	0	Yes (0.0.0.0/0 matches everything)	Not used, shorter match

Matching entry: 192.168.2.0/24. Next-hop router selected: 10.0.0.3. The default route also matches, but with prefix length 0 it loses to the /24.

3. Analysis – Routing Configuration Errors

The lookup logic is correct: the table as written already sends 192.168.2.100 to 10.0.0.3, and the destination is inside 192.168.2.0/24. So the problem is not the matching rule, and I list the faults that can still break delivery, which must be checked with the router:

- **Wrong or swapped next hop:** 10.0.0.3 may not be the router actually attached to 192.168.2.0/24 (for example it was swapped with 10.0.0.2). Check with **ping 10.0.0.3** and **show ip route**.
- **Next hop not reachable:** the table shows no connected route or exit interface for 10.0.0.0/24, so a static route whose next hop is not on a connected network is not installed.
- **No return route:** the router at 10.0.0.3 may lack a route back to the sender or to the 192.168.2.0/24 LAN (or its LAN interface is down), so replies are lost.
- **Incomplete notation:** the default is shown as "Default Route", not 0.0.0.0/0; no exit interface, metric or administrative distance is recorded, so the entries are hard to verify.

4. Optimization – Improved Routing Table

Destination	Next hop	Exit interface	Type	Purpose
10.0.0.0/24	directly connected	g0/0	C	Makes all next hops reachable
192.168.1.0/24	10.0.0.2	g0/0	S	Customer network 1
192.168.2.0/24	10.0.0.3	g0/0	S	Customer network 2 (verify the correct router)
0.0.0.0/0	10.0.0.1	g0/0	S*	Default route, gateway of last resort

- **Summarisation was considered but rejected:** 192.168.1.0/24 and 192.168.2.0/24 cannot be combined into one /23 (a /23 must start at an even third octet), and a /22 (192.168.0.0/22) would wrongly include 192.168.0.0 and 192.168.3.0. They also use different next hops, so they must stay as separate /24 routes.
- Specific routes are kept as /24 so longest-prefix match always beats the default route, and only unknown traffic goes to 10.0.0.1.

Route used if no destination matches

The default route 0.0.0.0/0 via next hop 10.0.0.1 (for example a packet to 8.8.8.8 or 192.168.3.5). It appears as the gateway of last resort.

5. Implementation (Cisco Packet Tracer) – commented configuration

```
Router(config)# interface g0/0
 Router(config-if)# ip address 10.0.0.254 255.255.255.0       ! connected link to 10.0.0.0/
24
 Router(config-if)# no shutdown
Router(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.2   ! network 1
Router(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.3   ! network 2
Router(config)# ip route 0.0.0.0 0.0.0.0 10.0.0.1             ! default route
```

6. Verification

- Packet Tracer: **show ip route** must show C 10.0.0.0/24, S 192.168.1.0/24 [1/0] via 10.0.0.2, S 192.168.2.0/24 [1/0] via 10.0.0.3 and S* 0.0.0.0/0 via 10.0.0.1 ("Gateway of last resort is 10.0.0.1"). Then **ping 192.168.2.100** and **tracert 192.168.2.100**; the first hop must be 10.0.0.3.
- Wireshark (filter **icmp or arp**) on the link to 10.0.0.3: ARP for 10.0.0.3 followed by ICMP echo request to 192.168.2.100 proves the packet used the correct next hop. A request with no reply points to the missing return route; a missing ARP reply points to a wrong next hop.

Conclusion: 192.168.2.100 belongs to 192.168.2.0/24, matches that entry by longest-prefix match, and goes to next hop 10.0.0.3. The table is logically correct, so the failure comes from next-hop reachability, a wrong or swapped next hop or a missing return route; the improved table with connected route, exit interface and 0.0.0.0/0 via 10.0.0.1 resolves and documents it.